

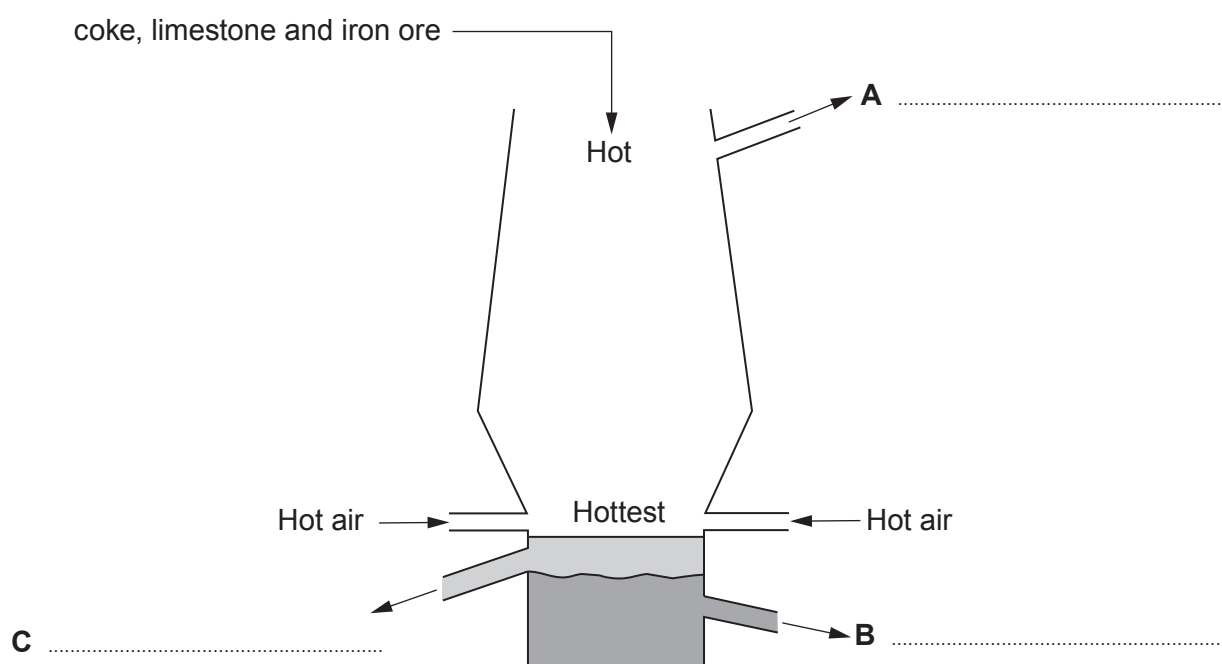
5. Port Talbot steel works in South Wales is one of the largest in the UK. It is capable of producing 5 million tonnes of steel every year.

A number of raw materials are used in a blast furnace to produce iron.

- (a) The diagram below shows the blast furnace and the substances that are added. Hot waste gases, molten iron and slag are the products formed.

Complete the labelling of the diagram.

[2]



- (b) The table below shows some information about the substances used in the blast furnace.

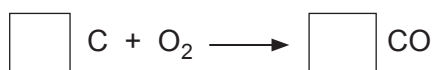
Complete the table.

[4]

Substance	Chemical name	Chemical formula	Type of material
iron ore		Fe_2O_3	compound
coke	carbon		
limestone	calcium carbonate	CaCO_3	

- (c) The blast furnace relies on the processes of reduction and oxidation.

Complete the balanced symbol equations for two of the processes occurring in the blast furnace. [3]



- (d) The table below shows the cost of the raw materials required to produce one tonne of iron in the blast furnace.

Raw material	Mass of raw material (tonnes)	Cost per tonne of raw material (£)	Total cost of raw material (£)
iron ore	1.75	80	
coke	0.25	120	30
limestone	0.25	60	15
air	4.00	3	12
Total cost to produce one tonne of iron			£

Complete the table above to calculate the total cost of materials required to produce one tonne of iron. [2]

- (e) Aluminium is also extracted from its ore but by a different process. Explain why aluminium cannot be extracted in a blast furnace using coke (carbon). [1]

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